WORKED SOLUTION

# STEP I 2014, Question 9

Mechanics • Projectile with Horizontal Force

**STEP I 2014, Question 9**

Read the question before you go on. It is in the STEP Support Programme's question database:

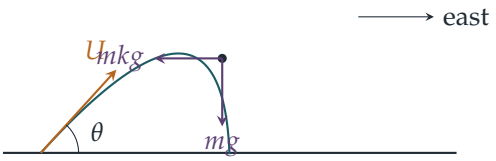
step.maths.org/questions/14-s1-q9

## The key idea

Both forces are constant, so the acceleration is constant in each direction and we can treat the horizontal and vertical motions separately, exactly as for an ordinary projectile. The vertical motion doesn't feel the westward force at all. The new feature is that the horizontal velocity is steadily reduced, so the particle stops moving east at some moment and then drifts back west. The whole shape of the path depends on *when* that moment comes compared with the top of the flight and the landing.

## Solution

Model the particle as a point mass moving in a vertical plane, with no forces other than its weight and the given horizontal force. Take $x$ east and $y$ vertically upwards, with the origin at the point of projection.




**The times $T_H$ and $T_L$**

Newton's second law in each direction, after dividing by $m$, gives

$$\ddot{x} = -kg, \qquad \ddot{y} = -g.$$

The initial velocity is $(U\cos\theta,\ U\sin\theta)$, so integrating twice,

$$\dot{x} = U\cos\theta - kgt, \qquad x = Ut\cos\theta - \tfrac{1}{2}kgt^2,$$
$$\dot{y} = U\sin\theta - gt, \qquad y = Ut\sin\theta - \tfrac{1}{2}gt^2.$$

The greatest height is reached when $\dot{y} = 0$, and the particle lands when $y = 0$ with $t > 0$:

$$\boxed{T_H = \frac{U\sin\theta}{g}, \qquad T_L = \frac{2U\sin\theta}{g}.}$$

These are the same as with no westward force, because that force has no vertical component.

#### The meaning of $T$

The given time $T = \dfrac{U\cos\theta}{kg}$ is exactly when $\dot{x} = 0$. Before $T$ the particle is moving east; after $T$ it moves west. So, if the particle is still in the air at time $T$, that is the moment when it is *furthest east*. Comparing the three times,

$$\frac{T_H}{T} = \frac{U\sin\theta}{g}\cdot\frac{kg}{U\cos\theta} = k\tan\theta, \qquad \frac{T_L}{T} = 2k\tan\theta.$$

It will also help to know where the particle lands:

$$x(T_L) = T_L\left(U\cos\theta - \tfrac{1}{2}kgT_L\right) = \frac{2U^2\sin\theta}{g}(\cos\theta - k\sin\theta) = \frac{2U^2\sin\theta\cos\theta}{g}(1 - k\tan\theta).$$

This is positive (east of the start) if $k\tan\theta < 1$ and negative (west) if $k\tan\theta > 1$.

**Ask yourself** The path is no longer symmetric, so what does it actually look like?

The acceleration $(-kg, -g)$ is constant, so the path is still a parabola, but with its axis tilted: it is the parabola we'd get under a "gravity" of $g\sqrt{1+k^2}$ pointing down and to the west. The top of the flight is no longer the turning point of the parabola, and the three cases below are about where the furthest-east point (time $T$) falls.

#### The three cases

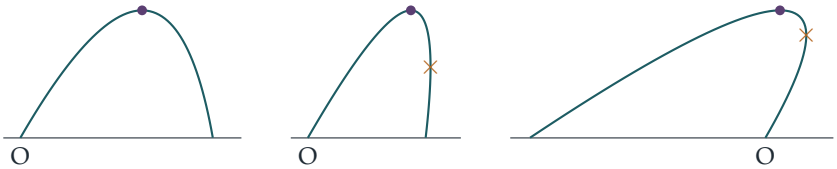
**Case $k\tan\theta < \frac{1}{2}$.** Then $T_H < T_L < T$. The particle lands before its eastward velocity has run out, so it moves east throughout. The path looks like an ordinary trajectory, rising to its highest point and then coming down more steeply than it went up, since the horizontal speed keeps falling. It lands east of the start.

**Case $\frac{1}{2} < k\tan\theta < 1$.** Now $T_H < T < T_L$. The particle passes its highest point while still moving east, carries on east while descending until time $T$, where its velocity is vertically downwards, and then curls back westwards before landing. Since $k\tan\theta < 1$ it still lands east of the start, but west of its furthest-east point.

**Case $k\tan\theta > 1$.** Now $T < T_H$. The eastward motion stops while the particle is still rising, at which point it is moving vertically upwards. It reaches its greatest height while already moving west and comes down to land *west* of the point of projection.

The sketches show the three cases with $\theta = 60^\circ$ and the same $U$ and $g$, with $k\tan\theta \approx$

0.35, 0.6 and 1.8 respectively. The dot marks the highest point, at $T_H$, the cross marks the furthest-east point, at $T$, and O is the point of projection.




$k\tan\theta < \frac{1}{2}$ $\qquad$ $\frac{1}{2} < k\tan\theta < 1$ $\qquad$ $k\tan\theta > 1$

**The case $k\tan\theta = 1$**

Now $T = T_H$, so at the top of the flight *both* velocity components vanish together and the particle is momentarily at rest. Also $x(T_L) = 0$, so it lands back at the point of projection.

In fact the motion is along a straight line. When $\tan\theta = 1/k$, the initial velocity $U(\cos\theta, \sin\theta)$ is a positive multiple of $(k, 1)$, while the acceleration $(-kg, -g) = -g(k, 1)$ points exactly the opposite way. A particle whose acceleration is constant and directly opposed to its velocity moves in a straight line: it travels up the line of projection, decelerating at $g\sqrt{1+k^2}$, stops at time $T_H$, and slides straight back down the same line to land where it started at time $T_L = 2T_H$. You can confirm this from the equations: with $k = \cot\theta$,

$$x\tan\theta = Ut\sin\theta - \tfrac{1}{2}gt^2 = y \quad \text{for all } t,$$

so the particle never leaves the line $y = x\tan\theta$.

## Checks

- Setting $k \to 0$ makes $T \to \infty$, so we are always in the first case, and $x(T_L)$ becomes the familiar range $U^2\sin 2\theta/g$.
- The landing position $\frac{2U^2\sin\theta\cos\theta}{g}(1 - k\tan\theta)$ changes sign exactly at $k\tan\theta = 1$, matching the boundary between the second and third cases.
- With $U = g = 1$, $\theta = 60°$ and $k = 0.45$, the particle lands at $x \approx 0.191$ but reaches $x = \frac{\cos^2\theta}{2k} \approx 0.278$ at time $T \approx 1.11$, between $T_H \approx 0.87$ and $T_L \approx 1.73$, so it does turn back before landing. A step-by-step numerical integration of the motion agrees with all three times and the landing point.

## Common slips

- Letting the westward force change the vertical motion: $T_H$ and $T_L$ don't involve $k$ at all.
- Assuming the highest point is the "middle" of the path. With the horizontal force

the path is a tilted parabola, so the highest point and the furthest-east point are different.

- Drawing the second case as if the particle lands west of the start: it turns back, but not far enough, since $k\tan\theta < 1$.

## Beyond STEP

Constant acceleration in any direction always gives a parabola. Here the effective gravity is simply tilted, and rotating the axes to line up with it turns the problem back into an ordinary projectile. That idea, choosing coordinates adapted to the forces, is central to first-year mechanics, as are charged particles in uniform electric fields, which behave in just this way. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

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